

SVILENGRAD, Bulgaria

WMO: 157410

Lat: 41.767N

Lon: 26.200E

Elev: 180

StdP: 14.60

Time Zone: 2.00 (EUE)

Period: 94-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.1	20.8	11.9	10.1	18.6	15.8	12.3	23.8	21.9	42.0	18.1	39.0	1.7	N/A	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	95.7	71.9	92.8	71.1	90.2	69.8	74.2	91.7	72.4	89.3	71.0	87.3	4.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	103.9	82.8	66.6	98.4	79.8	65.4	94.3	78.5	37.8	92.3	36.1	89.6	34.9	87.3	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	14.9	12.5	DB	7.6	102.4	7.0	3.9	2.6	105.2	-1.5	107.5	-5.5	109.6	-10.6	112.5
			WB	8.6	77.5	5.6	1.3	4.6	78.5	1.3	79.3	-1.8	80.0	-5.8	81.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	56.5	36.4	40.7	46.4	54.5	64.3	72.5	77.4	76.1	67.8	57.7	46.2	36.9	(6)
(7)		DBStd	16.00	7.22	7.74	7.42	6.93	5.72	4.91	4.24	4.59	5.94	7.38	7.55	9.63	(7)
(8)		HDD50	1488	424	268	165	37	1	0	0	0	0	17	158	418	(8)
(9)		HDD65	4268	886	680	578	315	80	6	1	0	40	248	564	870	(9)
(10)		CDD50	3865	3	8	52	174	446	674	850	810	535	255	45	13	(10)
(11)		CDD65	1169	0	0	0	2	60	230	385	345	125	21	1	0	(11)
(12)		CDH74	12899	0	0	11	35	763	2532	4386	3814	1179	176	3	0	(12)
(13)		CDH80	5612	0	0	2	2	210	1019	2153	1822	363	41	0	0	(13)
(14)	Wind	WSAvg	3.5	3.1	4.0	4.4	4.3	3.4	3.6	4.1	3.5	3.1	3.0	2.7	3.2	(14)
(15)	Precipitation	PrecAvg	21.9	1.5	1.6	1.7	1.8	2.2	1.5	1.7	1.2	1.7	2.0	2.4	2.6	(15)
(16)		PrecMax	33.9	4.8	4.9	4.3	3.1	6.1	4.2	6.5	2.2	5.4	4.2	4.6	5.4	(16)
(17)		PrecMin	17.3	0.1	0.3	0.2	0.4	0.6	0.1	0.0	0.0	0.0	0.2	0.6	1.2	(17)
(18)		PrecStd	4.4	1.3	1.2	1.1	0.8	1.5	1.0	1.5	0.6	1.5	1.3	1.3	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.9	65.3	74.9	78.2	90.3	95.6	102.0	100.1	92.1	84.9	71.6	63.3	(19)
(20)		MCWB	53.0	54.0	59.4	61.1	69.2	70.9	72.1	73.1	68.4	66.1	61.1	56.1	(20)	
(21)		2%	DB	54.6	61.0	67.4	74.6	85.1	91.7	96.2	95.6	88.0	78.5	66.4	59.6	(21)
(22)		MCWB	49.4	51.6	55.6	59.5	66.1	70.7	72.6	71.7	67.3	64.7	57.9	54.3	(22)	
(23)		5%	DB	50.0	57.9	63.9	71.1	81.4	88.8	93.2	92.7	84.5	74.1	62.3	55.2	(23)
(24)		MCWB	46.2	49.8	53.7	58.1	64.9	69.7	71.6	70.9	66.4	62.8	56.3	51.3	(24)	
(25)		10%	DB	46.2	53.8	60.1	67.6	78.0	85.9	90.4	89.5	81.0	70.5	58.1	50.9	(25)
(26)		MCWB	43.9	47.5	51.2	56.5	63.2	68.1	70.2	69.4	65.0	60.3	53.7	48.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	54.7	59.9	62.8	71.2	74.0	76.6	76.7	70.1	67.5	61.2	56.6	(27)
(28)		MDCB	59.3	63.4	71.8	74.7	86.9	90.3	94.8	96.0	84.8	80.8	69.2	62.3	(28)	
(29)		2%	WB	50.2	52.5	56.7	60.6	67.6	72.1	74.7	73.9	68.4	65.7	59.1	54.6	(29)
(30)		MDCB	54.4	60.0	66.5	71.2	81.6	88.3	92.9	91.5	83.2	76.7	64.5	59.2	(30)	
(31)		5%	WB	46.8	50.1	54.2	59.1	65.8	70.8	72.7	72.0	67.3	64.0	56.9	51.7	(31)
(32)		MDCB	49.0	56.9	62.8	68.7	79.1	86.4	89.6	89.2	81.6	73.2	62.5	54.3	(32)	
(33)		10%	WB	44.0	47.8	52.0	57.6	64.2	69.2	71.2	70.4	65.9	61.8	53.8	48.3	(33)
(34)		MDCB	46.0	53.6	59.0	66.2	76.3	83.8	87.8	86.7	78.9	69.0	57.6	50.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	12.1	18.1	19.4	19.6	23.8	24.7	24.5	24.7	22.5	18.8	15.2	12.1	(35)	
(36)		5% DB	MCDBR	19.0	25.7	25.4	25.9	30.6	29.8	28.8	29.6	27.6	24.6	22.2	16.2	(36)
(37)		MCWBR	14.2	16.8	14.8	13.0	14.6	12.5	10.5	11.2	11.0	12.1	14.1	11.7	(37)	
(38)		5% WB	MCDBR	17.1	24.2	23.6	22.2	27.8	27.7	26.7	26.8	24.6	22.2	19.6	14.7	(38)
(39)		MCWBR	13.6	16.3	14.1	11.9	14.0	12.7	11.1	11.0	10.4	11.5	13.6	11.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.349	0.380	0.422	0.472	0.449	0.449	0.449	0.456	0.436	0.404	0.375	0.355	(40)	
(41)		taud	2.336	2.245	2.143	2.015	2.161	2.198	2.211	2.187	2.212	2.284	2.315	2.313	(41)	
(42)		Ebn at Noon	248	257	259	254	264	263	262	256	252	245	235	233	(42)	
(43)		Edh at Noon	28	35	43	53	47	45	44	44	40	33	28	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	509	762	1118	1522	1907	2078	2164	1949	1442	940	592	447	(44)	
(45)		RadStd	64	81	97	161	129	162	129	90	103	93	63	64	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.85	N/A	N/A	N/A	+1.04	+1.12	-160	-232	+166	N/A

Nomenclature: See separate page